PROBLEM DA8

Lift of a Flapping Fly Wing

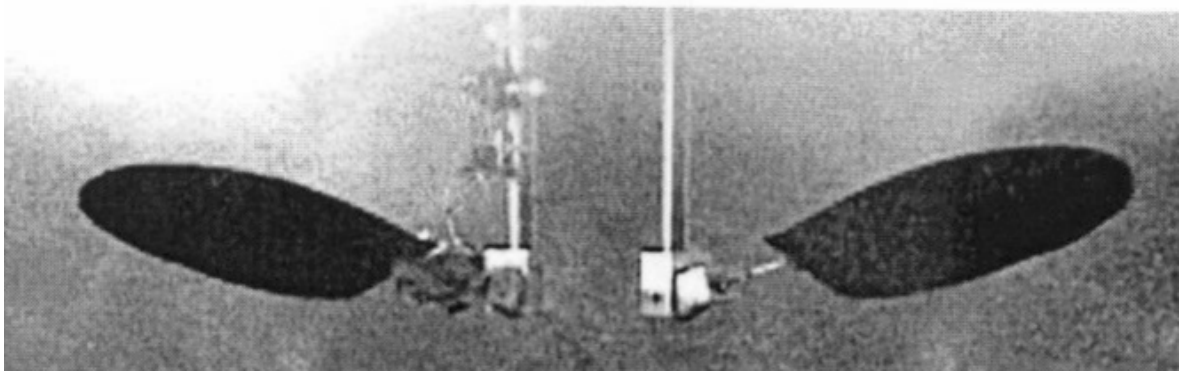
Original statement · DA8.pdf

Π theorem | Reynolds similarity | Experimental table

OFFICIAL PROBLEM STATEMENT

A paper published in Scientific American in 2001 by Prof. Michael Dickinson describes an experimental facility designed to determine the lift L generated by a fly suspended at a fixed point by the flapping motion of its wings of length L at a frequency f. The facility includes a large container filled with mineral oil. A pair of wings, geometrically similar to those of the fly, is placed within the container. Programmable electrical engines move the wings, reproducing the wing motion of the fly at varying frequencies. To mimic the characteristic fly dynamics, the wings in the experiment are made out of a very light material, so that their weight and inertia are entirely negligible. To guide the experiment design and analyze the results follow the steps indicated below.

The sheet uses L both for the lift and for the wing length; from question 1 on the lift is called F, as below.



Photograph of the sheet (crop of DA8.pdf): the pair of robotic wings of Dickinson's facility, which flap in a tank of mineral oil

1. Use dimensional analysis to express the lift force F as a function of the minimum number of dimensionless parameters.

2. Assuming the following characteristic values for the fly: flapping frequency $f = 200$ Hz, wing length $L = 2.5$ mm, air density $\rho = 1.2$ kg/m^3, air viscosity $\mu = 1.8 \times 10^{-5}$ kg/m·s, along with the nominal characteristics of the experiment: flapping frequency $f_e = 0.2$ Hz, wing length $L_e = 250$ mm, mineral oil density $\rho_e = 800$ kg/m^3, determine the mineral-oil viscosity μ_e needed to guarantee that the experiment is a physically similar representation of the fly dynamics.

3. The experimental facility including the model with $L_e = 250$ mm and the mineral oil selected above is used to investigate the variation of the lift with the flapping frequency. To that end, the wings are moved at different frequencies f_e and the corresponding forces F_e are measured, yielding the data points shown in the table below. Generate a curve for the variation of the force F with the flapping frequency f for the fly of wing length $L = 2.5$ mm.

f_e **(Hz)**	F_e **(N)**
0.14	1.33
0.15	3.33
0.16	4.80
0.17	7.13
0.18	8.00
0.19	6.60
0.20	5.07
0.21	4.53

`Source: DA8.pdf (page 1, exam of Dec 16, 2011, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; physical similarity, Eqs. 6.57-6.59.`

Worked solution

Hypotheses, Data and Dimensional Table

- The wings have no weight and no inertia of their own, and the fly is held at a fixed point: the lift depends only on the wing motion (frequency and size, the stroke shape fixed by geometric and kinematic similarity) and on the fluid.
- No free surface and negligible weight: no gravity.

Parameter	**Meaning**	**Dimensions**
F	lift (unknown)	MLT^{-2}
f	flapping frequency	T^{-1}
L	wing length	L
ρ	fluid density	ML^{-3}
μ	fluid viscosity	$ML^{-1}T^{-1}$

Application of the Π Theorem

$$F = F(f, L, \rho, \mu)$$

$n = 4$; (f, L, ρ) have independent dimensions (rows $f(0,0,-1)$, $L(1,0,0)$, $\rho(-3,1,0)$, $\det = -1 \cdot (1 \cdot 1 - 0 \cdot (-3)) = -1 \neq 0$ expanding along the first row), so one group remains.

Force. $[F] = [\rho]^a [f]^b [L]^c$: $M:\ 1 = a$; $T:\ -2 = -b$; $L:\ 1 = -3 + c$; so $\rho f^2 L^4$.

Viscosity. $[\mu] = [\rho]^a [f]^b [L]^c$: $M:\ 1 = a$; $T:\ -1 = -b$; $L:\ -1 = -3 + c$; so $\rho f L^2$, and the group is $\rho L^2 f / \mu$, a Reynolds number with the velocity scale fL.

$$\frac{F}{\rho L^4 f^2} = \Pi\left(\frac{\rho L^2 f}{\mu}\right)$$

Viscosity of the Oil

Physical similarity requires equal Reynolds numbers:

$$\frac{\rho L^2 f}{\mu} = \frac{\rho_e L_e^2 f_e}{\mu_e} \implies \mu_e = \mu \, \frac{\rho_e}{\rho} \left(\frac{L_e}{L}\right)^2 \frac{f_e}{f}$$

$$\mu_e = 1.8 \times 10^{-5} \times \frac{800}{1.2} \times \left(\frac{250}{2.5}\right)^2 \times \frac{0.2}{200} = 1.8 \times 10^{-5} \times 666.7 \times 10^4 \times 10^{-3} = 0.12 \ \mathrm{kg/(m\ s)}$$

Lift Curve of the Fly

Each measured point, with $\mu_e = 0.12$ kg/(m s), is similar to the fly state with the same Reynolds number:

$$\frac{\rho L^2 f}{\mu} = \frac{\rho_e L_e^2 f_e}{\mu_e} \implies f = \frac{\rho_e}{\rho}\left(\frac{L_e}{L}\right)^2 \frac{\mu}{\mu_e} f_e = 666.7 \times 10^4 \times 1.5 \times 10^{-4}\, f_e = 1000\, f_e$$

and then equal dimensionless lift:

$$\frac{F}{\rho L^4 f^2} = \frac{F_e}{\rho_e L_e^4 f_e^2} \implies F = \frac{\rho}{\rho_e}\left(\frac{L}{L_e}\right)^4 \left(\frac{f}{f_e}\right)^2 F_e = \frac{1.2}{800} \times 10^{-8} \times 10^6\, F_e = 1.5 \times 10^{-5}\, F_e$$

The same factor follows in the key's form: substituting f/f_e from the first relation, $F = (\rho_e/\rho)(\mu/\mu_e)^2 F_e = 666.7 \times (1.5 \times 10^{-4})^2 F_e = 1.5 \times 10^{-5} F_e$.

f_e (Hz)	F_e (N)	f (Hz)	F (N)
0.14	1.33	140	2.0×10^{-5}
0.15	3.33	150	5.0×10^{-5}
0.16	4.80	160	7.2×10^{-5}
0.17	7.13	170	1.07×10^{-4}
0.18	8.00	180	1.20×10^{-4}
0.19	6.60	190	0.99×10^{-4}
0.20	5.07	200	7.6×10^{-5}
0.21	4.53	210	6.8×10^{-5}

Answers and Checks

- Reduced law: $F/(\rho L^4 f^2) = \Pi(\rho L^2 f/\mu)$.
- Oil viscosity: $\mu_e = 0.12$ kg/(m s).
- Fly: $f = 1000 f_e$ and $F = 1.5 \times 10^{-5} F_e$; maximum lift 1.2×10^{-4} N at 180 Hz.
- Dimensions: $\rho L^4 f^2$ is $ML^{-3} \cdot L^4 \cdot T^{-2} = MLT^{-2}$, a force.

- Order of magnitude: a fly of about 10 mg weighs $\sim 10^{-4}$ N, the order of the maximum lift, which is reasonable for a hovering insect.
- Reynolds number of the fly: $\rho L^2 f/\mu = 1.2 \times 6.25 \times 10^{-6} \times 200/1.8 \times 10^{-5} \simeq 83$, neither small nor large: inertia and viscosity both matter, so no limit simplifies the law and the experiment must match Re exactly.
- Check against the official key: DA8.pdf gives $\mu_e = 0.12$ kg/(m s) and the table f = 140 to 210 Hz with F = 0.2×10^{-4}, 0.5×10^{-4}, 0.72×10^{-4}, 1.07×10^{-4}, 1.2×10^{-4}, 1×10^{-4}, 0.76×10^{-4}, 0.68×10^{-4} N, the same values.